A computable omega-to-omega barrier with progressive free sets that compute the halting set

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Editorial note. Solutions in the counted statement are infinite f-free sets, as in the progressive free-set principle1023-1026 and the proof1072. Finite free sets are not claimed to compute the jump.

Editorial note. Normalize the modulus to be nondecreasing and at least3, which is harmless and ensures all finite thin-coloring instances lie in the proved n>=2 range.

Editorial note. The finite-jump coloring array is shared support of the existing Schreier omega-jump task and is not counted again; the new object is a computable progressive coloring on a non-computably-bounded barrier.

Editorial scope. One original result. Full source-local mathematical construction included; named classical inputs are explicit.

Allowed standard prerequisites (editorial).

- Classical computability tools. The halting set has a monotone left-c.e. modulus; the uniform relativized limit lemma removes one jump by one extra coordinate; a function dominating a modulus computes the set.
- Classical finite-order barrier characterization. Assous PropositionII.1: a barrier has order type at most omega^n exactly when its member sizes are bounded by n. The source omega-bounded equivalence is supplied as local proof support.

There exist four other subsystems which, together with RCA_0, calibrate the strength of most theorems. These subsystems are known as the "Big Five" (see Montálban [2]). Among these, ACA_0 (Arithmetic Comprehension Axiom) extends the axioms of RCA_0 with the comprehension scheme for all arithmetic formulas. Based on the correspondence between computability and definability, it is equivalent to stating the existence of the Turing jump of any set ($\forall X \exists Y (Y = X')$).

We shall also consider two lesser-known stronger variants of ACA_0, namely, ACA_0' and ACA_0^+. The system ACA_0' extends RCA_0 with the axiom stating closure under all finite jumps; the system ACA_0^+ extends RCA_0 with the axiom stating the existence of the ω-jump of any set. The ω-jump of a set X is the set $X^{(\omega)} = \bigoplus_n X^{(n)}$. The system ACA_0^+ is famous for being the best known upper bound to Hindman's theorem.

Let us first fix some notation and recall the definition of these principles for colorings of finite subsets of fixed size. We denote by $\mathbb{N}$ the set of natural numbers and by $\mathbb{N}^+$ the set of positive natural numbers. For $X \subseteq \mathbb{N}$ and $n \in \mathbb{N}$ we denote by $[X]^n$ the set of all n-subsets of X. We denote by $[X]^{<\omega}$ the set of all finite subsets of X. We always assume that sets are presented in increasing order and make no distinction between a set and the sequence of its elements in increasing order. We write $x \cdot y$ for the concatenation of two sequences x and y. We identify a natural number with the set of its predecessors, so that, if $k \geq 1$ we can write $f : X \to k$ to declare a function from X to $\{0, 1, \ldots, k-1\}$. A function of this type is often called a coloring of X in k colors.

Definition 0.1. *Let C be any non-empty set. Given a coloring $f : [\mathbb{N}]^n \to C$, a set $H \subseteq \mathbb{N}$ is* f-thin for color $c \in C$ *if $c \notin f([H]^n)$. A set $H \subseteq \mathbb{N}$ is f-thin (or thin for f) if H is f-thin for some color $c \in C$.*

The definition of f-thin set depends on the choice of codomain of the function. It will always be clear from the context.

Definition 0.2. *A set $B \subseteq [\mathbb{N}]^{<\omega}$ is a* barrier *if*

(1) $\mathrm{base}(B)$ *is infinite;*

(2) for every $X \in [\mathrm{base}(B)]^\omega$, there is some $s \in B$ such that $s \preceq X$;

(3) for every $s, t \in B$, $s \not\subset t$.

Given a set X and $n \in \mathbb{N}$, we write $[X]^{\leq n}$ for the set of all subsets $s \subseteq X$ such that $|s| \leq n$, and $[X]^{\leq !\omega}$ for the set of all subsets $s \subseteq X$ such that $|s| \leq 1 + \min s$. By convention, for $n = 0$, $[X]^{\leq n}$ is the singleton $\{\emptyset\}$. Given a function $h : \mathbb{N} \to \mathbb{N}$, we write $[X]^{\leq h(\cdot)}$ for the set of all finite $s \subseteq X$ such that $|s| \leq h(\min s)$.

Definition 0.3. *A set $B \subseteq [\mathbb{N}]^{<\omega}$ is* ω-bounded *if and only if $B \subseteq [\mathbb{N}]^{\leq h(\cdot)}$ for some function $h : \mathbb{N} \to \mathbb{N}$. It is* computably ω-bounded *if furthermore h is computable.*

A function $h : \mathbb{N} \to \mathbb{N}$ is *left-c.e.* if there is a uniformly computable sequence of functions $h_0, h_1, \ldots$ such that for every $x, i \in \mathbb{N}$, $h_i(x) \leq h_{i+1}(x)$ and $\lim_i h_i(x) = h(x)$. The sequence $(h_i)_{i \in \mathbb{N}}$ is then called a *left-c.e. approximation* of h. If B is a computable barrier of order type at most ω^ω, then it is ω-bounded by a left-c.e. function. This bound is tight, as there exist computable barriers of order type ω^ω which are not computably ω-bounded.

Definition 0.4. *Let $B \subseteq [\mathbb{N}]^{<\omega}$ be a set and $f : B \to [\mathbb{N}]^{<\omega}$ be a coloring. A set $H \subseteq \mathrm{base}(B)$ is* f-free *if for every $s \in B \cap [H]^{<\omega}$, $f(s) \cap H \subseteq s$.*

Definition 0.5. *Fix $B \subseteq [\mathbb{N}]^{<\omega}$. A function $f : B \to [\mathbb{N}]^{<\omega}$ is* progressive *if for every $s \in B$, either $f(s) = \emptyset$, or $\min f(s) \geq \min s$.*

An easy combinatorial argument shows that the following statement is classically true.

Definition 0.6 (Progressive Free Set Theorem)**.** *Fix a set $B \subseteq [\mathbb{N}]^{<\omega}$ and a bounding function $b : \mathbb{N} \to \mathbb{N}$. PFS_b^B is the statement "For every b-constrained progressive coloring $f : B \to [\mathbb{N}]^{<\omega}$, there is an infinite f-free set".*

Proposition 0.7. *There exists a computable barrier B of order type ω^ω and a computable progressive coloring $f : B \to \mathbb{N}$ such that every f-free set computes $\emptyset'$.*

Lemma 0.8. *There exists two computable arrays $(e_{n,k})_{n,k\in\mathbb{N}}$ and $(d_{n,k})_{n,k\in\mathbb{N}}$ of Turing indexes such that for every $n, k \in \mathbb{N}^+$ with $n \geq 2$, $\Phi_{e_{n,k}}^{\emptyset^{(k)}}$ is a coloring $f_n^k : [\mathbb{N}]^n \to n$ such that for every infinite f_n^k-thin set H, $\Phi_{d_{n,k}}^{H} = \emptyset^{(k)}$.*

Proof. Let $k \in \mathbb{N}^+$. Let g_k be a uniform modulus of the set $\emptyset^{(k)}$. Note that Δ^0_{k+1}-indexes for each function g_k can be found computably, uniformly in k.

For $n \geq 2$ let $f_n^k(x_0, \dots, x_{n-1}) = n-1$ if $g_k(x_0) \leq x_1$ and if $g_k(x_0) > x_1$, let $f_n^k(x_0, \dots, x_{n-1}) = i$ for $i < n-1$ the largest value such that $g_k(x_0) > x_{i+1}$.

Let H be an infinite f_n^k-thin set for some color c. The color c cannot be $n-1$ as for a given $x_0 \in H$ there exists $x_1 < \dots < x_{n-1} \in H \setminus \{0, \dots, g_k(x_0)\}$, hence $f_n^k(x_0, \dots, x_{n-1}) = n-1$.

We claim H is thin for color $n-2$. Indeed, assume by contradiction that there exists some tuple $x_0 < \dots < x_{n-1} \in H$ such that $g_k(x_0) > x_{n-1}$ so that f_n^k takes color $n-2$ on $\{x_0, \dots, x_{n-1}\}$. Since $c < n-2$ and H is infinite, there exists some $y_{c+2} < \dots < y_{n-1} \in H \setminus \{0, \dots, g_k(x_0)\}$, therefore $f_n^k(x_0, \dots, x_{c+1}, y_{c+2}, \dots, y_{n-1}) = c$, contradicting our assumption that H is f_n^k-thin for c.

Write $H = \{x_0 < x_1 < \dots\}$. For every $i \in \mathbb{N}$, $g_k(x_i) \leq x_{i+n-1}$, hence H computes a function dominating g_k and therefore computes $\emptyset^{(k)}$. This computation can be done uniformly in the set H, k and n. □

Lemma 0.9. *There exists a functional Γ such that for every set X and every index e, if $\Phi_e^{X'}$ is a coloring $f : [\mathbb{N}]^n \to \ell$, then $\Gamma^X(e)$ is a coloring $g : [\mathbb{N}]^{n+1} \to \ell$ such that if an infinite set H is g-thin for a color c, then H is f-thin for the same color c.*

Proof. Consider a set X and an index e such that $\Phi_e^{X'}$ is a coloring $f : [\mathbb{N}]^n \to \ell$ for some $n, \ell \in \mathbb{N}^+$. Let $(f_s)_{s\in\mathbb{N}}$ be a $\Delta^0_2(X)$-approximation of f (indexes for such an approximation, where the f_s are seen as X-computable functions, can computably be found uniformly in X and e). Finally, consider the following X-computable coloring g defined by $g(x_0, \dots, x_n) = f_{x_n}(x_0, \dots, x_{n-1})$ (again, the construction is uniform).

Let $H \subseteq \mathbb{N}$ be an infinite set. If for some $x_0 < \dots < x_{n-1} \in H$ we have $f(x_0, \dots, x_{n-1}) = c$ for some color c, then $f_s(x_0, \dots, x_{n-1}) = c$ for every s bigger than a certain threshold. Thus, as H is infinite, there exists some $x_n \in H$ such that $g(x_0, \dots, x_n) = c$. □

Combining these two lemmas, we get the following, where the array $(d_{n,k})_{n,k\in\mathbb{N}}$ is as in Theorem 0.8.

Lemma 0.10. *There exists a uniformly computable sequence of colorings $f_{n,k} : [\mathbb{N}]^{n+k} \to n$, for $n, k \in \mathbb{N}^+$, such that for every infinite $f_{n,k}$-thin set H, $\Phi_{d_{n,k}}^{H} = \emptyset^{(k)}$.*

Lemma 0.11. *A barrier B has order type at most ω^ω if and only if B is ω-bounded.*

Proof. Suppose first $B \subseteq [\mathbb{N}]^{\leq h(\cdot)}$ for some function $h : \mathbb{N} \to \mathbb{N}$. For every $s \in B$, let $\alpha_s = \sum_{i<|s|} \omega^{h(\min s)-i} s(i)$. Note that $s <_{\text{lex}} t$ if and only if $\alpha_s < \alpha_t$, so B has order type at most ω^ω since $\alpha_s < \omega^\omega$ for every $s \in B$.

Suppose now B has order type at most ω^ω. Let $x \in \mathbb{N}$ and $B_x = \{s : x \cdot s \in B\}$. Then B_x is a barrier of order type at most ω^{n_x} for some $n_x \in \mathbb{N}^+$. By Assous [1, Proposition II.1], a barrier B has order type at most ω^n if and only if $B \subseteq [\mathbb{N}]^{\leq n}$, so $B_x \subseteq [\mathbb{N}]^{\leq n_x}$. Let $h(x) = n_x$. Then $B \subseteq [\mathbb{N}]^{\leq h(\cdot)}$. □

Proof. Let $g : \mathbb{N} \to \mathbb{N}$ be the modulus of $\emptyset'$ and let $(g_n)_{n\in\mathbb{N}}$ be a left-c.e. approximation of g. Without loss of generality, it can be assumed that $g_n(x) \geq 1$ for every $x, n \in \mathbb{N}$ and that g_n is non-decreasing for each $n \in \mathbb{N}$.

Let B be defined as follows: for $x \in \mathbb{N}$ and $s \subseteq \mathbb{N}$ with $x < \min s$, let $x \cdot s \in B$ if and only if $|s| = g_{\min s}(x)$.

We claim that B is a barrier with base $\mathbb{N}$: for any infinite subset $X = \{x_0, x_1, \dots\}$ of $\mathbb{N}$, we have $\{x_0, x_1, \dots, x_{g_{x_1}(x_0)}\} \in B$ (since $g_{x_1}(x_0) \geq 1$). Let $x \cdot s, y \cdot t$ be in B and suppose $x \cdot s \subseteq y \cdot t$. Then $x \geq y$ and $\min s \geq \min t$. Moreover $|s| = g_{\min s}(x) \geq g_{\min s}(y) \geq g_{\min t}(y) = |t|$. Thus $s = t$ and $x = y$. By Lemma 0.11, the order type of B is ω^ω.

Let $f : B \to \mathbb{N}$ be defined by $f(x \cdot s) = x + h_{|s|-1}(s) + 1$ where $h_n : [\mathbb{N}]^{n+1} \to n$ is the computable instance of TS^{n+1}_n obtained in Lemma 0.10 such that every infinite h_n-thin set computes $\emptyset'$. Let H be an infinite f-free set. For every $x \in H$, there exists some bound $b_x > g(x)$ such that $g_{b_x}(x) = g(x)$. For every $s \subseteq H \setminus [0, b_x]$ of cardinality $g(x)$, we have $f(x \cdot s) = x + h_{g(x)-1}(s) + 1$. There are two cases:

Case 1: $H \setminus [0, b_x]$ is $h_{g(x)-1}$-thin for some $x \in H$, in that case, by definition of $h_{g(x)-1}$, $H \geq_T \emptyset'$.

Case 2: $H \setminus [0, b_x]$ is not $h_{g(x)}$-thin for every $x \in H$, in that case, for every $x \in H$ and every $y \in (x, x + g(x))$, there exists some $s \subseteq H \setminus [0, b_x]$ such that $f(x \cdot s) = y$ and therefore, since H is f-free, y is not in H. So $H \geq_T \emptyset'$ as the principal function of H dominates g. □

References

[1] Marc Assous. Caractérisation du type d'ordre des barrières de Nash-Williams. *Publ. Dép. Math. (Lyon)*, 11(4):89–106, 1974.

[2] Antonio Montalbán. Open questions in reverse mathematics. *Bulletin of Symbolic Logic*, 17(03):431–454, 2011. Publisher: Cambridge Univ Press.